

Appendices

Appendix A - Phonological Scansion Procedure

In this section, I will discuss the theoretical Phonological Scansion procedure as derived from the conclusions of Hayes and Kaun (1996), and then the practical process derived from it and implemented in the Calliope application (as well as the patterns the process uses to match meters). I also detail some applications of the procedure in other languages and in solving some metrical problems as a demonstration of its effectiveness and validity.

A1 Theoretical Process

1. Mark the stress contours on a word/compound (0 – 5+)
2. Mark the relative stresses within the word (ws)
3. Mark the following syllables for Metrical Strength (the value the syllable receives with reference to the metrical scheme)

Type	Stress Level	Metrical Strength	Confidence Level
Polysyllabic compound/ phrasal verbs	Strong	Strong	High
Polysyllabic non-compound	Strong	Strong	Moderate
End of Phonological Phrase	Strong	Strong	Very High
End of Intonational Unit	Strong	Strong	Very High

Table 50: Phonological Scansion phonological units

4. Deduce the metrical schemes appropriate from:
 - a. the position of the strong stresses,
 - b. syllable number
 - c. metrical schemes of other lines
5. Match strong metrical syllables to the metrical scheme of strong stresses (where weak stresses may be strong or weak)
6. Match the remaining syllables, rationalizing any elided syllables

A2 Practical Process

The following process describes the Phonological Scansion procedure. It is partly implemented in the Calliope application. Steps which are not implemented in the application are coloured blue and marked.

1. Get Stresses
 - a. Mark the stresses in the line

- b. Determine the relative stresses in each PP (w=weak, n=some stress, m=subordinate strong stress, s=strong stress)
 2. Record Key Stresses
 - a. Record all the stresses in a polysyllabic word (PW)
 - b. Record the left-most stress in a Clitic Phrase (CP)
 - c. Record the left-most stress and its preceding stress in a Phonological Phrase (PP)
 - d. Record the left-most stress and its preceding stress in an Intonational Unit (IU)
 3. Get the Syllable Count
 - a. Count the syllables
 - b. Match the stresses to patterns with the syllable count, else resolve missing or combined stresses and recount the syllables and get the new matching pattern.
 - c. Final weak stresses
 - i. Final weak stresses may be feminine endings i.e./ extra stress on pattern
 - ii. Two final weak stresses are very likely to be double feminine endings (wn or ww)
 - d. Resolve stresses
 - i. resolve ww>W, or nw->N
 - ii. stress clashes (ss, ms) > s-s,
 - iii. add additional stress at intonational breaks or phonological phrase breaks
 4. Record type of matches
 - a. Find the number and type of stresses that match each pattern (allocate the greatest unit to the stress pattern where there is more than one unit).
 5. Evaluate the matches
 - a. Evaluate which pattern matches best – give greater priority to final units, then IU, then PP, then PW, then CP.
 Give the matches the following numerical values and choose the pattern with the highest total:
 IU=3
 PP=2
 PW3+ (a polysyllabic word with 3 or more syllables)=2
 CP=1
 PW2 (a polysyllabic word with 2 syllables)=1
 For triple rhythms:
 - i. allocate patterns which do not match a triple rhythm to a None pattern, e.g./ mws, swm, sw, s)
 - ii. assign dactyls and anapaests with strongly marked dipping rhythm (swm and mws) to None.
 - iii. less marked dipping rhythms (e.g./ swn, smw or wms, nws) are permissible anapaests and dactyls
- (NOTE: the following steps are not implemented in the Calliope application)
- b. Where there is a tie,

- i. use surrounding patterns.
 - ii. If the syllable count is low, the line may be in no known meter.
- c. Where the line is ambiguously trochaic or iambic, use the rhythm of polysyllabic words to determine the meter – the rhythm of the greatest number of words establishes the meter:
 - i. Disyllabic words with a ws pattern are iambic, sw are trochaic
 - ii. Use the pattern of the final two syllables of polysyllabic words
- d. Accentual verse can be identified where the number of strong stresses in a line is constant, even if the number of syllables changes. There is likely also to be no dominant rhythmical pattern in the line.
- e. Free verse can be identified where there is no dominant rhythmical pattern, and the number of strong stresses varies.

A2.1 Example

Line: Shall I compare thee to a summer's day?

Stresses: Shall(m) I(n) com(w)pare(s) thee(w) to(n) a(w) sum(s)mer's(w) day(s)?

Phonological Analysis: <{[/Shall I compare\ thee]} {[to a /summer's\] [day?]}>

Key Stresses: <{[/mnws\w]}{[xx/sw\]s]}>

Phonological Scansion Analysis:

10 syllables, 4 main stresses

Total duple units to match=1IU, 1PW2, 2PW4 = 8

Iambic Units (ws) = final IU, 1PW2, 1PW4 (total=6)

Trochaic Units (sw) = 1PW4 (total=2)

Spondaic Units (ss) = none

Pyrrhic Units (ww) = none

Total triple units to match=2PW4, 1PW2 = 5

Anapaestic Units (wws) = none

Amphibrachic Units (wsw) = 1PW2 (total=1)

Dactylic Units (sww) = 1PW4, 1PW4 (total=4)

The line matches only 2/3 units with a dactylic rhythm, and is not therefore dactylic.

The line is iambic with 6/8 units matched (75%)

The analysis indicates that the line is iambic pentameter with the scansion mn/ws/wx/xs/ws. The iambic rhythm is very strong (75% regular).

A3 Patterns

Syllable Count	Meter Name	metrical scheme
2	Iambic	Ws
2	Trochaic	Sw
2	Spondaic	Ss
2	Pyrrhic	Ww
3	Anapaestic	Wws
3	Dactylic	SwW
3	Amphibrachic	Wsw
7	Anacreontic	None
11/11/11/5	Sapphic	Swsxswswswss (x3) Swswss
11/11/9/10	Alcaic	s/sw/ss/sww/sx/s (x2) s/sw/ss/sw/sx sww/sww/sw/sx
Variable (13-17)	Classical Dactylic Hexameter	Sw(w)/sw(w)/s,w(,w)/sw(w)/sww/sw (final feet= sww/sw, caesura in 3 rd foot)

Table 51: Phonological Scansion metrical patterns

A4 Application to Other Languages

One method of assessing the validity of Phonological Scansion in English poetry is to determine whether it works in similar contexts in other languages. In the section below I have applied the theory to accentual syllabic poetry in four languages, German, Neo-Aramaic, Coptic and Medieval Latin. However, it could also be applied to Dutch, Early Latin, Estonian, Modern Hebrew, Modern Greek, Old English, Old Icelandic and Norse dróttkvætt, Slovene and Russian, among others.

A4.1 German

A4.1.1 Analysis

German prosody is very much like English: it is derived from Classical Greek and Latin examples and applied to a stress-based language, resulting in accentual-syllabic meters.